

Fundamentals of Solid State Physics

Electronic Properties - The Free Electron Model

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Last Class

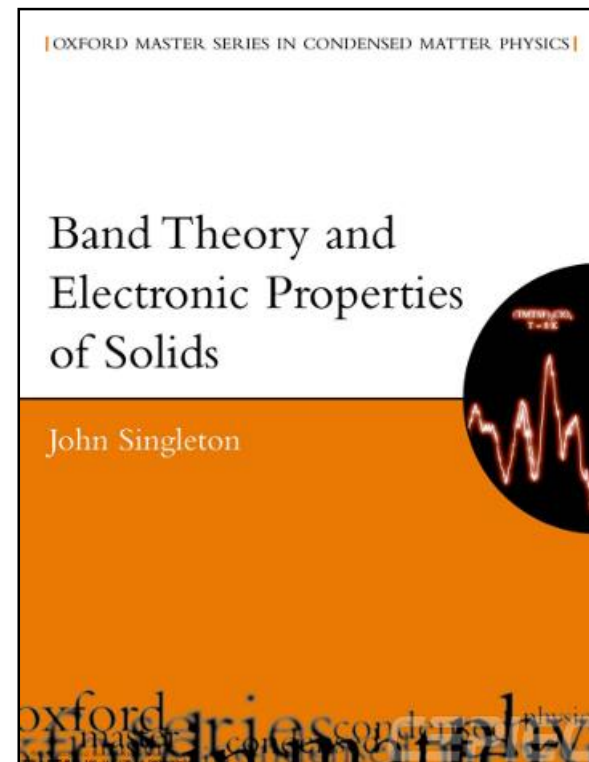
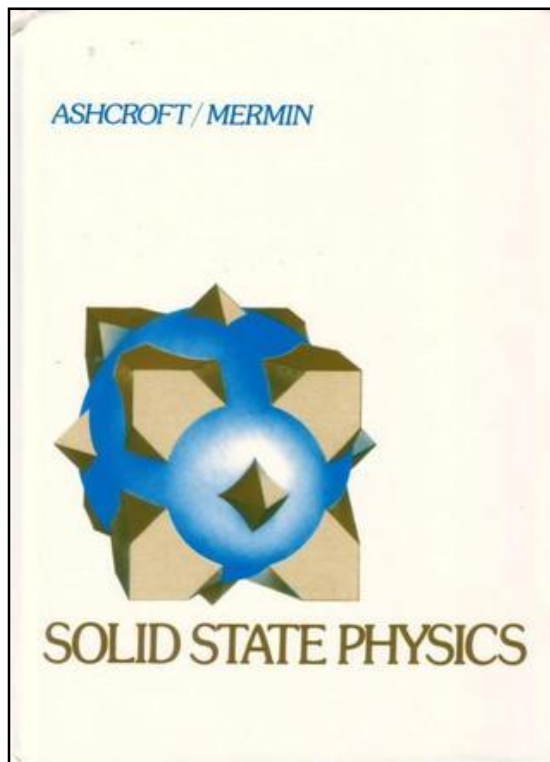
- Introduction (Week 1)
- **Materials and Crystal Structures (Week 2–3)**
 - **Crystal structures, lattices**
 - **Reciprocal space, Brillouin zones**
 - **Materials Characterization: Wave diffraction, the Bragg law**
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Further Reading

- Ashcroft & Mermin, Chapter 1, 2, 3
- Singleton, Chapter 1



Note about “E” and “e”

- Sorry, we run out of English letters...

$$E, \mathcal{E}$$

energy
(J or eV)

$$\mathbf{E}, \vec{E}$$

electric field
(V/m)

$$e$$

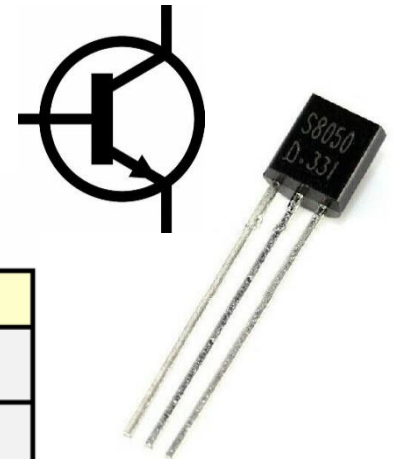
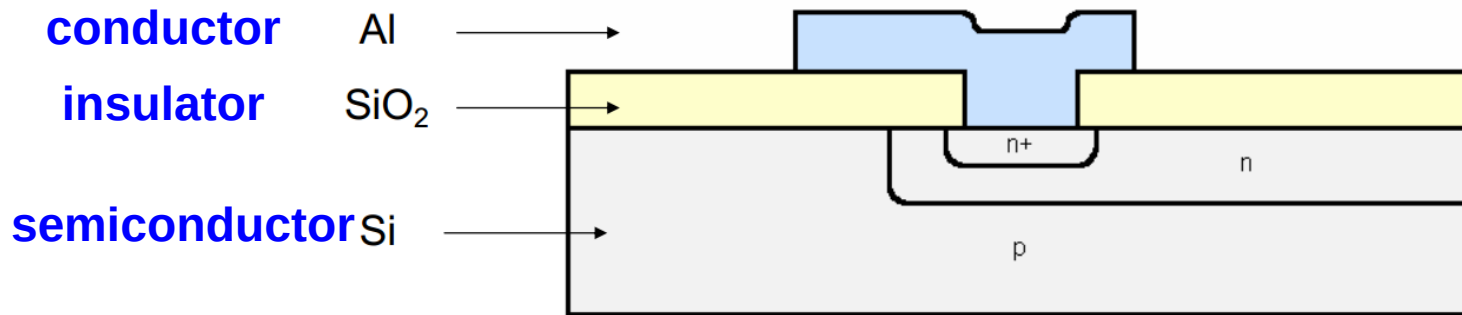
electron charge (q), $= 1.6 \times 10^{-19}$ C

Euler's number, $= 2.71828...$

Electronic Properties of Materials

CMOS transistor

- Complementary **Metal-Oxide-Semiconductor**



Metal



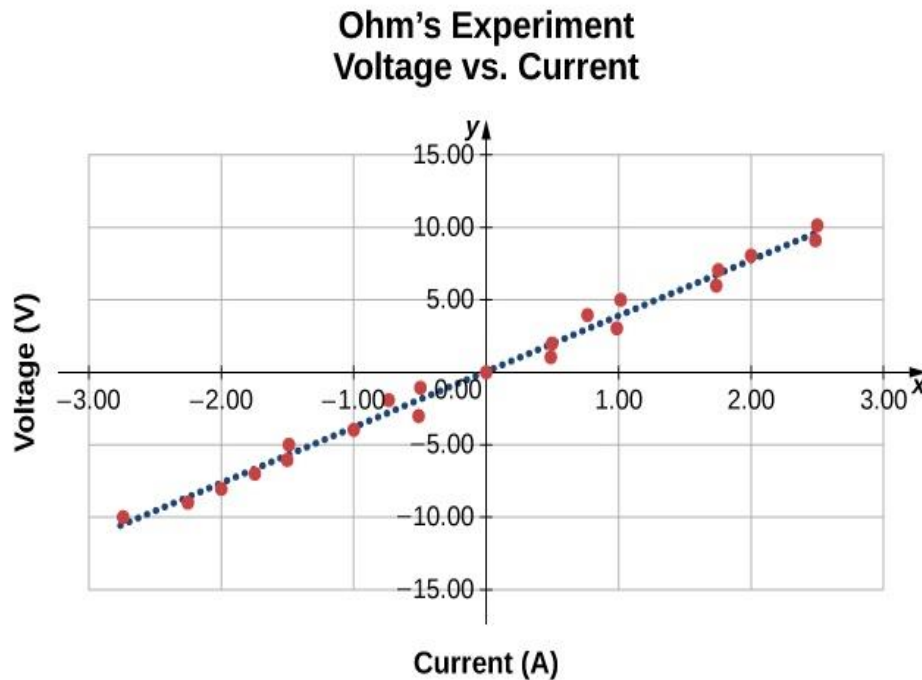
SiO₂



Silicon

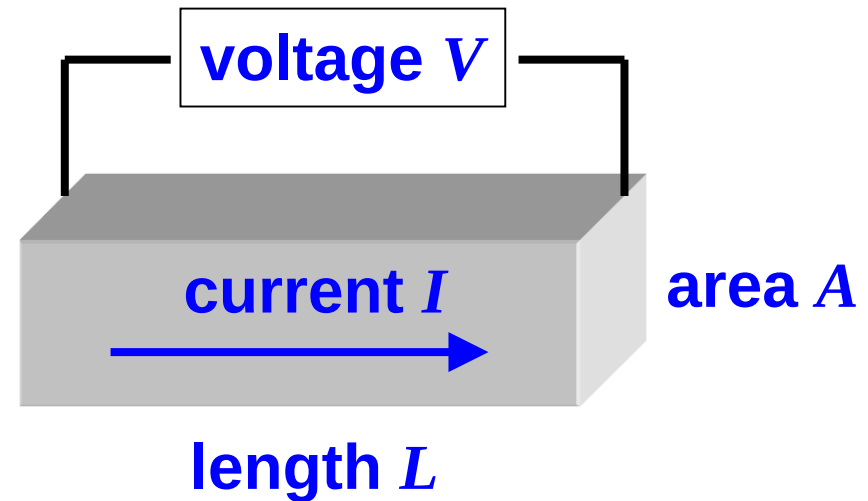
Electronic Properties of Metals

■ Ohm's Law (1827)



$$V = IR$$

$$R = 3.84\Omega$$



j - current density (A/m^2)

E - electric field (V/m)

ρ - resistivity ($\Omega\cdot\text{m}$)

σ - conductivity (S/m)

Electronic Properties of Metals

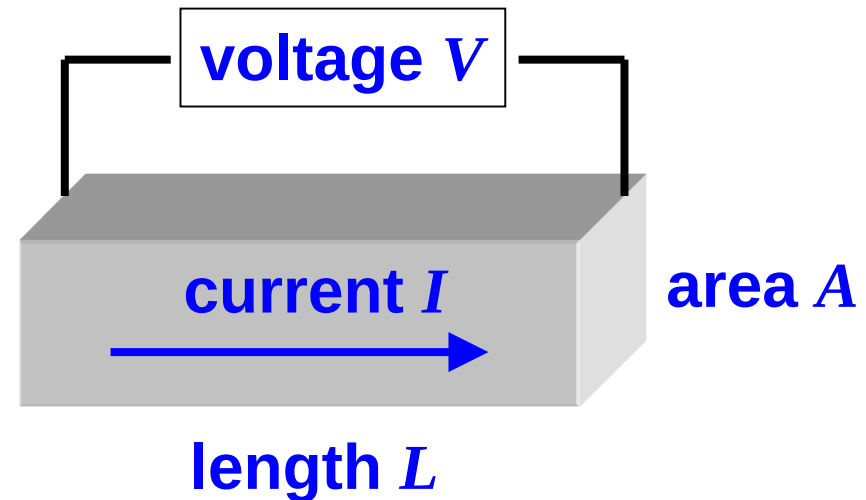
■ Ohm's Law (1827)

$$V = IR$$

macroscopic

resistance

$$R = \rho \frac{L}{A}$$



$$j = \frac{I}{A} = \frac{1}{\rho} \frac{V}{L} = \frac{1}{\rho} E$$

$$j = \sigma E$$

microscopic

j - current density (A/m^2)

E - electric field (V/m)

ρ - resistivity ($\Omega \cdot m$)

σ - conductivity (S/m)

Electronic Properties of Metals

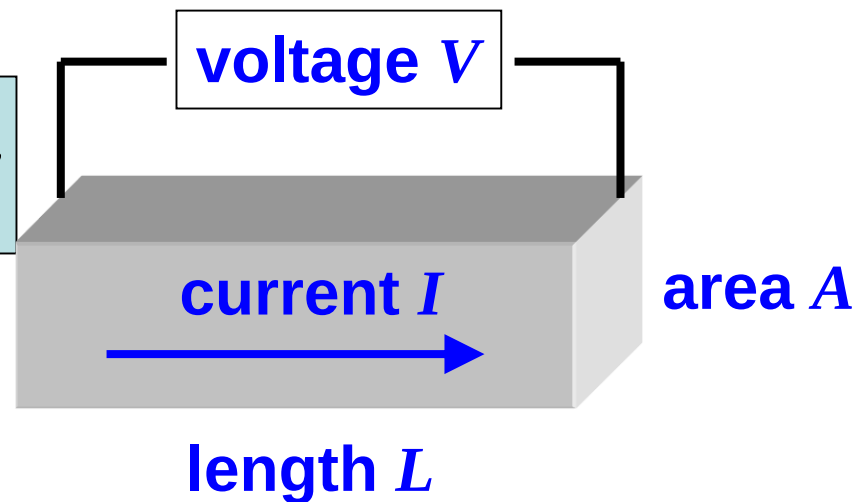
■ conductivity and mobility

$$I = \frac{\partial Q}{\partial t} = ne \frac{\partial(AL)}{\partial t} = neA \frac{\partial L}{\partial t} = neAv$$

$$j = \frac{I}{A} = nev$$

$$\sigma = \frac{\mathbf{j}}{\mathbf{E}} = ne \frac{\mathbf{v}}{\mathbf{E}} = ne\mu$$

$$\mu = \frac{\mathbf{v}}{\mathbf{E}}$$



n - density of carriers (#/m³)

v - velocity of carriers (m/s)

μ - carrier mobility (m²/V/s)

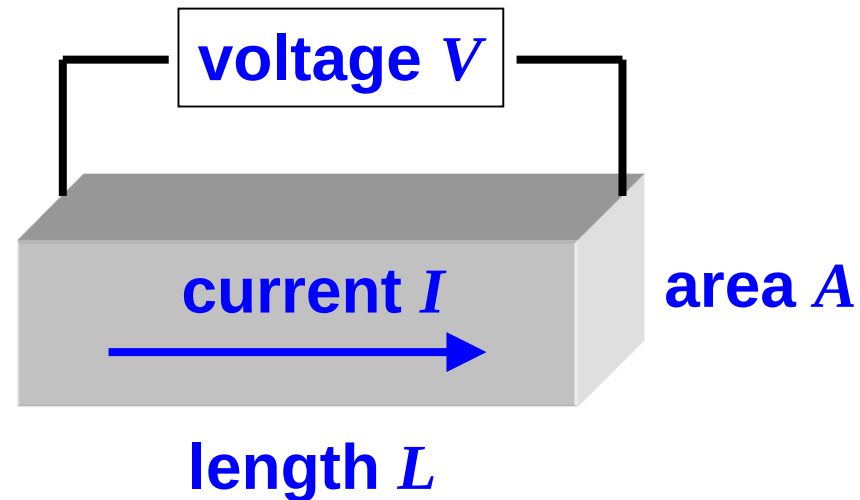
How to get Ohm's Law?

- assume free electrons in vacuum

$$\mathbf{F} = m\mathbf{a} = e\mathbf{E}$$

acceleration

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}}{m} = \frac{e\mathbf{E}}{m}$$



$$\mathbf{v} = \frac{e\mathbf{E}}{m}t \rightarrow \infty$$



and

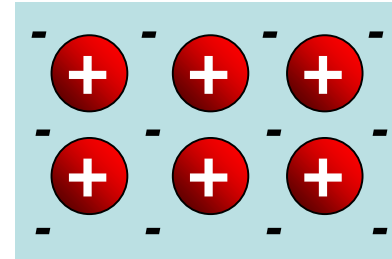
$$\sigma = ne \frac{\mathbf{v}}{\mathbf{E}} \rightarrow \infty$$



what is wrong?

The Drude Model 德鲁德模型

Free electron 'gas' model, in 1900



positive ions
+
electron cloud

- Independent
 - electrons do not interact with each other
- Free
 - electrons do not interact with ions, except collision
- Collision (Origin of the resistance)
 - electrons are scattered by the ions instantaneously
- Relaxation time τ 弛豫时间
 - average time between two collisions
 - electron mean free path $l = v^* \tau$
- Maxwell–Boltzmann distribution
 - average kinetic energy

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

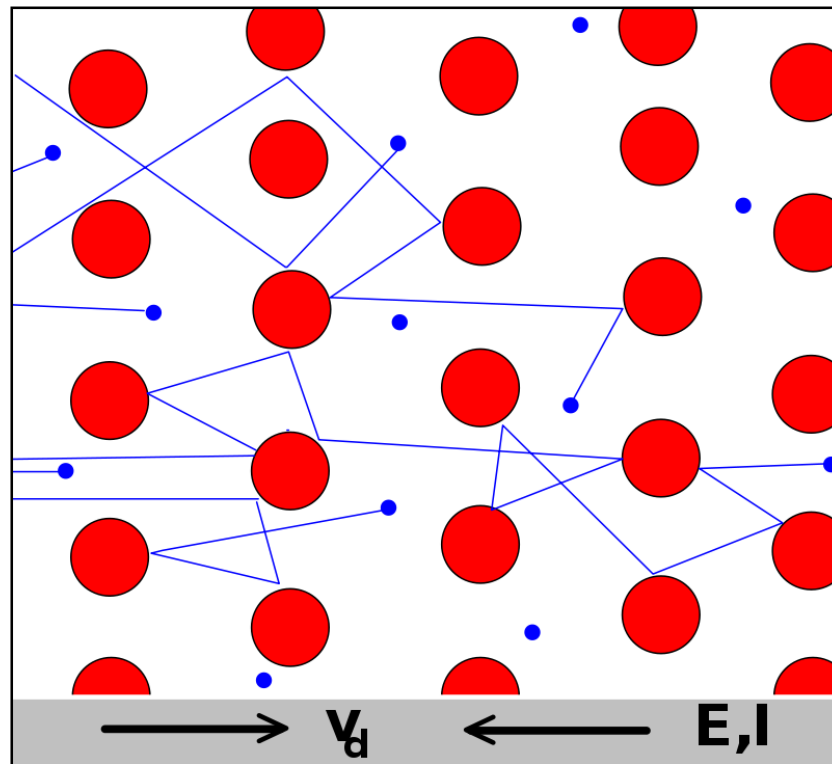


P. Drude
1863–1906

The electron was discovered
by J.J. Thompson in 1897

The Drude Model 德鲁德模型

Free electron 'gas'



Resistance is caused by *collisions* of electrons with atoms

The Drude Model 德鲁德模型

Drude-Lorentz Model

(Ashcroft & Mermin, p.11)

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt} + m \frac{\mathbf{v}}{\tau} = e\mathbf{E}(t)$$

τ - relaxation time (s) 弛豫时间

when E is constant, v is constant



$$\frac{d\mathbf{v}}{dt} = 0$$



$$m \frac{\mathbf{v}}{\tau} = e\mathbf{E}$$

$$\mu = \frac{\mathbf{v}}{\mathbf{E}} = e \frac{\tau}{m}$$

mobility

$$\sigma = ne\mu = ne^2 \frac{\tau}{m}$$

conductivity

$$\mathbf{j} = ne\mathbf{v} = \sigma\mathbf{E}$$

Ohm's law

Successes of The Drude Model

- Ohm's Law

$$\mathbf{j} = \sigma \mathbf{E}$$

- Electronic conductivity σ

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

$$v \sim 10^5 \text{ m/s}$$

$$\tau = \frac{l}{v}$$

$$\tau \sim 10^{-14} \text{ s}$$

$$\sigma = ne^2 \frac{\tau}{m}$$

$$\sigma \sim 10^7 \text{ S/m}$$

m = electron mass $9.11 \cdot 10^{-31} \text{ kg}$

$k_B = 1.38 \cdot 10^{-23} \text{ J/K}$

$e = 1.6 \cdot 10^{-19} \text{ C}$

$T = 300 \text{ K}$, room temperature

l = mean free path $0.1 \sim 1 \text{ nm}$

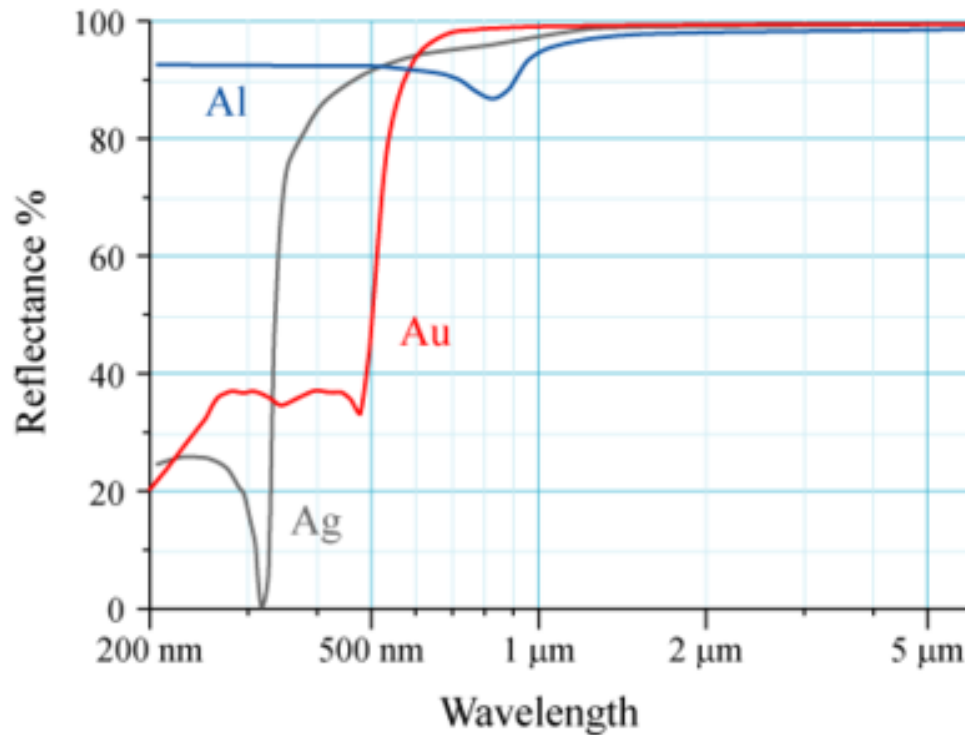
n = atomic density $\sim 10^{29} \text{ \#/m}^3$

metals	conductivity (S/m) at 300 K
Ag	$6.3 \cdot 10^7$
Cu	$6.0 \cdot 10^7$
Al	$3.5 \cdot 10^7$

Successes of The Drude Model

■ Optical Reflectivity of Metals

Lecture note 5.3



mirror reflection

Failures of the Drude Model

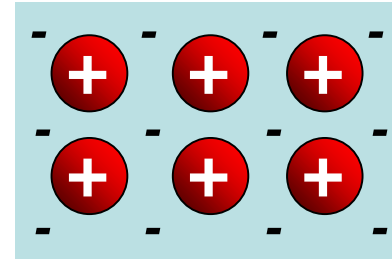
It cannot explain

- Electronic heat capacity
- Thermal conductivity
- Hall effect / Hall coefficient
- Insulators / Semiconductors
- ...

What was wrong?

The Drude Model 德鲁德模型

Free electron 'gas'



positive ions
+
electron cloud

- Independent
 - electrons do not interact with each other
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$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$



P. Drude
1863–1906

The Sommerfeld Model 索末菲模型

Free electron 'Fermi' gas (in 1928)

- Introduce **quantum mechanics**

- **Schrodinger Equation** (in 1925)

- ~~■ **Maxwell–Boltzmann distribution**~~

→ **Fermi–Dirac distribution** (in 1926)



A. Sommerfeld
1868–1951

Sommerfeld was nominated for the Nobel Prize 84 times, but got zero.

Many of his students won Nobel Prizes

The Electron Wave Function

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \cdot \psi(\mathbf{r}) = E\psi(\mathbf{r})$$

free electron

$$V(\mathbf{r}) = 0$$



$$\nabla^2 \psi(\mathbf{r}) = -k^2 \psi(\mathbf{r})$$

$$k^2 = \frac{2mE}{\hbar^2}$$



$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} A_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$\int_V \psi^* \cdot \psi d\mathbf{r} = 1$$



one solution is

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

The Electron Wave Function

energy

$$E(k) = \frac{\hbar^2 k^2}{2m}$$

velocity

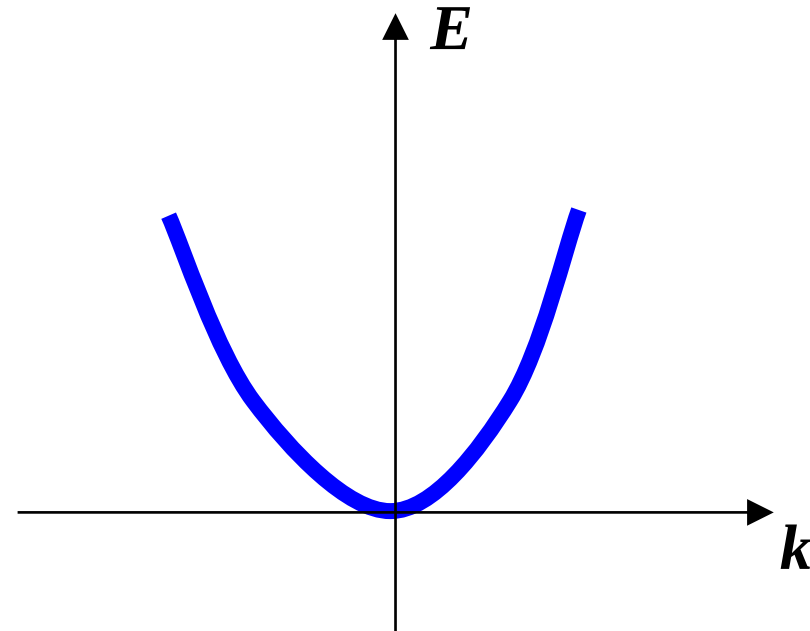
$$v = \frac{\hbar k}{m} = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

momentum

$$p = \hbar k$$

electron mass

$$\frac{1}{m} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$



E - k diagram
(energy dispersion curve)

Classical vs. Quantum

Classical

velocity

$$v$$

momentum 动量

$$p = mv$$

energy 能量

$$E = \frac{p^2}{2m}$$

Quantum

wavenumber 波数
wavevector 波矢

$$k = \frac{2\pi}{\lambda}$$

$$p = \hbar k$$

$$E = \frac{\hbar^2 k^2}{2m}$$

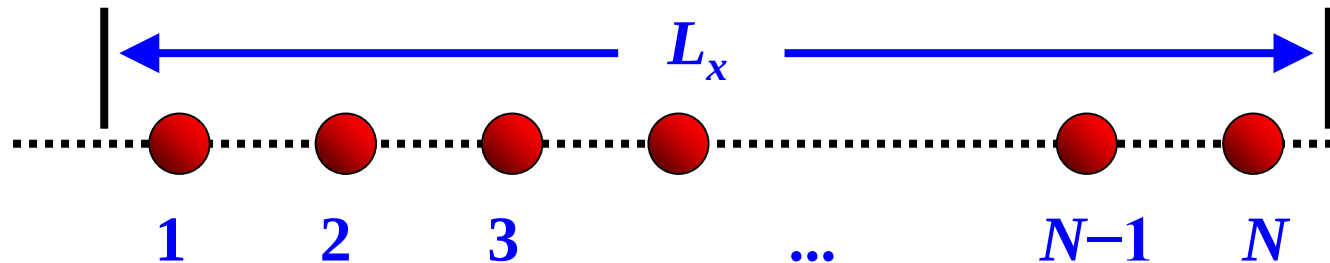
1D atomic chain

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} \exp(i\mathbf{k} \cdot \mathbf{r})$$



1D atomic chain

$$\psi(x) = \frac{1}{\sqrt{L_x}} \exp(ik_x x)$$



N is large $\sim 10^{23}$

Born-von Karman *periodic* boundary condition:

If two atoms are far apart ($N \gg 1$) in a crystal, they are exactly the same

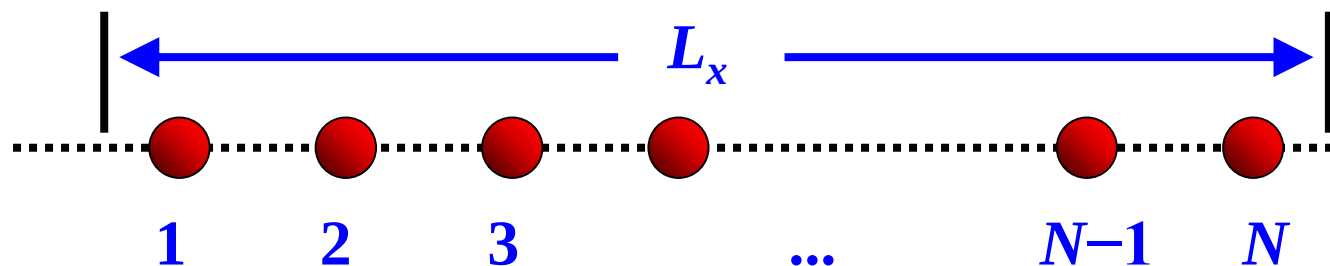
1D atomic chain

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} \exp(i\mathbf{k} \cdot \mathbf{r})$$



1D atomic chain

$$\psi(x) = \frac{1}{\sqrt{L_x}} \exp(ik_x x)$$



N is large $\sim 10^{23}$

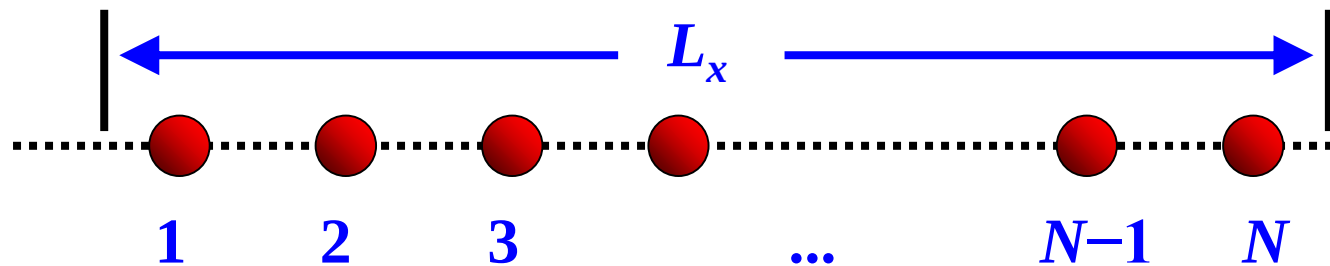
Born-von Karman *periodic* boundary condition

$$\psi(x) = \psi(x + L_x)$$



$$\exp(ik_x L_x) = 1$$

1D atomic chain



N is large $\sim 10^{23}$

Born-von Karman *periodic* boundary condition

$$\psi(x) = \psi(x + L_x) \quad \longrightarrow \quad \exp(ik_x L_x) = 1 \quad \longrightarrow \quad k_x L_x = 2\pi n_x$$

$$\longrightarrow \quad k_x = \frac{2\pi n_x}{L_x} \quad n_x = 0, \pm 1, \pm 2, \dots$$

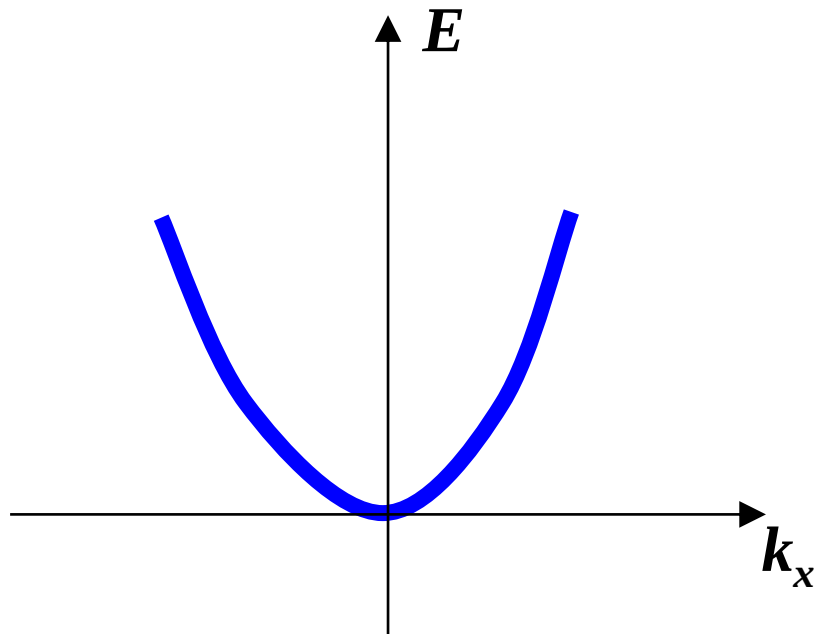
k is a *quantized* value

1D atomic chain

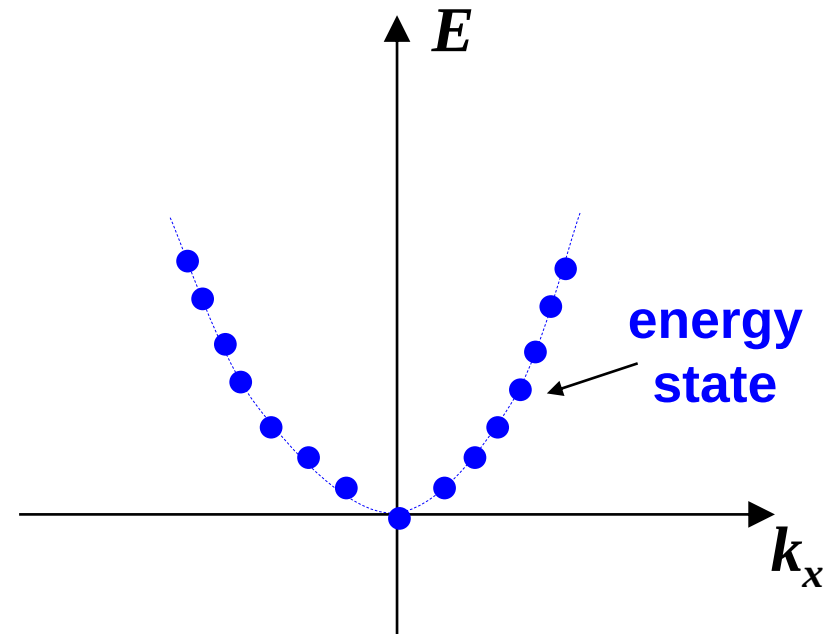
energy

$$E(k) = \frac{\hbar^2 k^2}{2m}$$

$$k_x = \frac{2\pi n_x}{L_x} \quad n_x = 0, \pm 1, \pm 2, \dots$$



E-k diagram
(energy dispersion curve)



quantized energy
state / level

1D atomic chain

energy

$$E(k) = \frac{\hbar^2 k^2}{2m}$$

$$k_x = \frac{2\pi n_x}{L_x}$$

$$n_x = 0, \pm 1, \pm 2, \dots$$

velocity

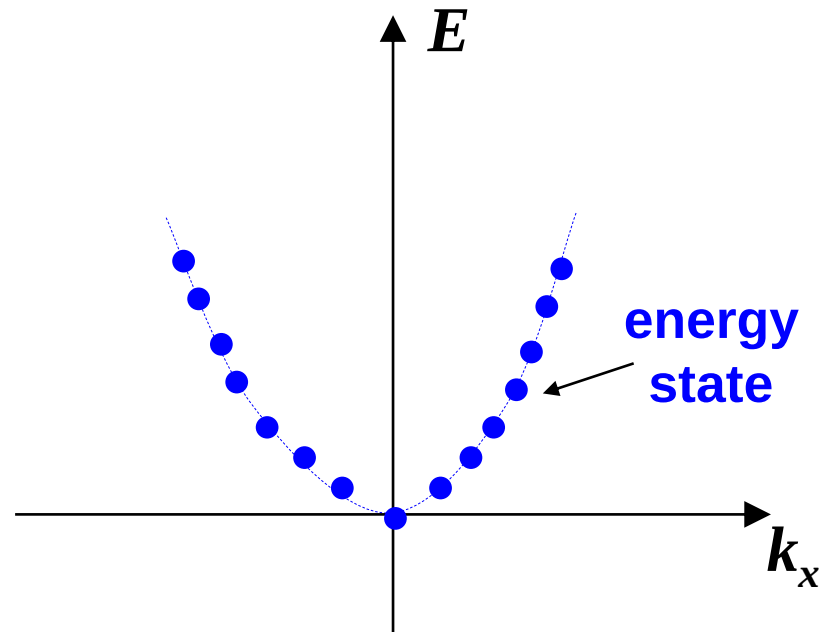
$$v = \frac{\hbar k}{m} = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

momentum

$$p = \hbar k$$

electron mass

$$\frac{1}{m} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$



When L is large,
 k and $E(k)$ are **quasi-continuous**
 (准连续)

**quantized energy
 state / level**

State vs. Electron

energy state / level / orbital
能态 / 能级 / 轨道



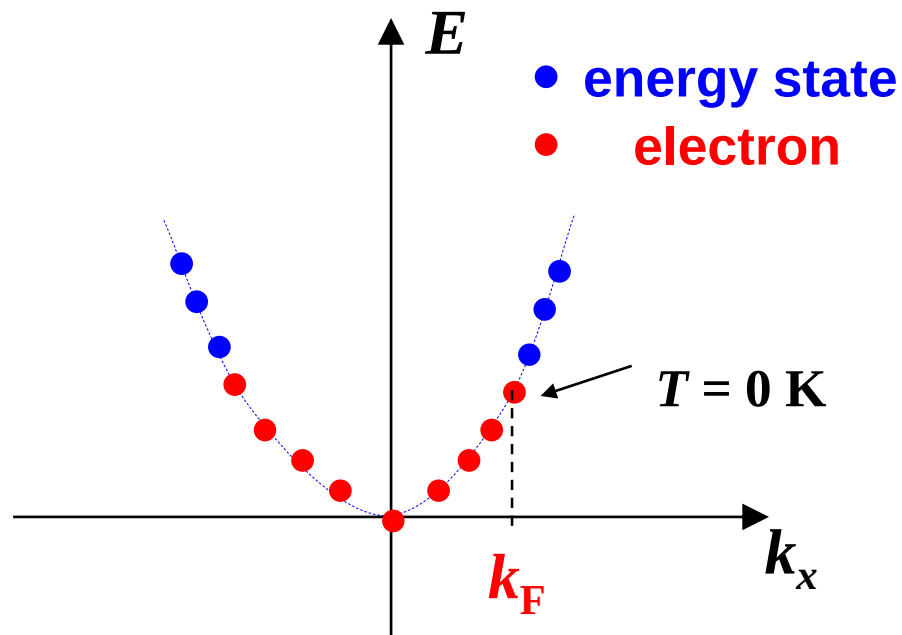
determined by space, lattice,
environments, ...

electron / phonon / ...
电子 / 声子 / ...



**There are infinite number
of states, but only finite
number of electrons.**

1D atomic chain: How many electrons?



$$k_x = \frac{2\pi n_x}{L_x} \quad n_x = 0, \pm 1, \pm 2, \dots$$

k_F - Fermi wavevector
highest **occupied** state at $T = 0 \text{ K}$

Electrons have high energy ($v > 0$)
even at $T = 0 \text{ K}$

Fermi-Dirac distribution:
spin up and down

$$-k_F < k_x < +k_F$$

$$N = 2 \cdot \frac{2k_F}{2\pi / L_x}$$

$$k_F = \frac{\pi}{2} \frac{N}{L_x}$$

N - total number of free electrons

N/L - free electron density

1D atomic chain

$$N = 2 \cdot \frac{2k_F}{2\pi / L_x}$$

1D atomic chain

$$k_F = \frac{\pi}{2} \frac{N}{L_x}$$

$$v_F = \frac{\hbar k_F}{m_e}$$

k_F - Fermi wavevector
 E_F - Fermi energy
 v_F - Fermi velocity
 T_F - Fermi temperature

$$E_F = \frac{\hbar^2 k_F^2}{2m_e}$$

$$T_F = \frac{E_F}{k_B}$$

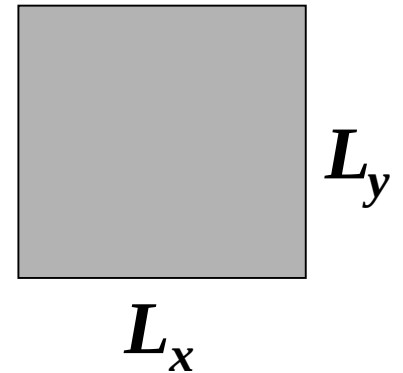
These values are determined by electron density N/L , not N or L

highest *occupied* state at $T = 0$ K

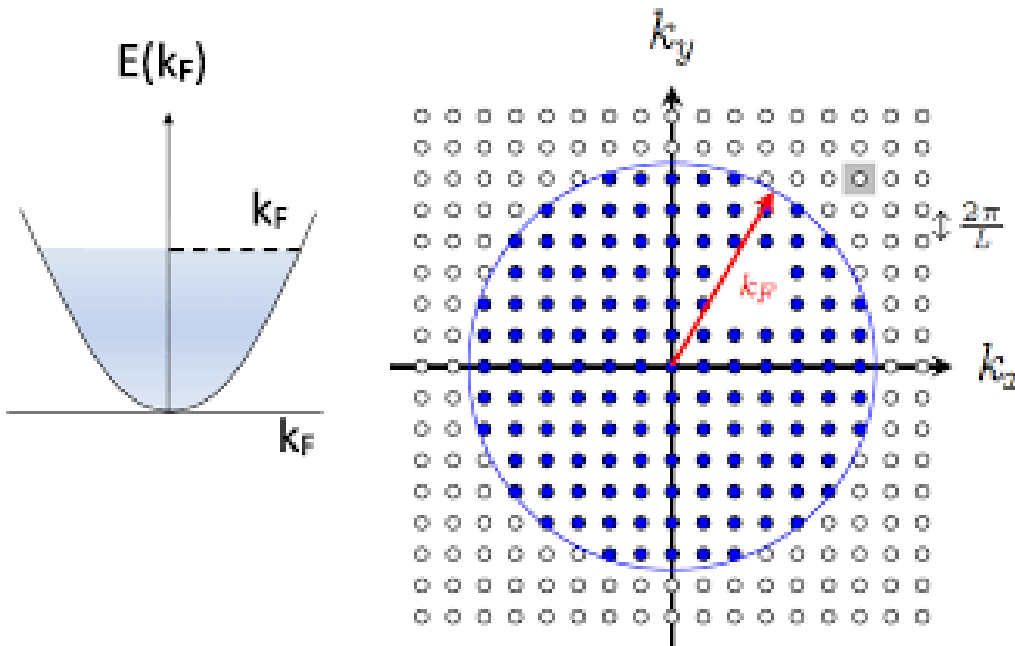
2D box

in a 2D box

$$k_x = \frac{2\pi n_x}{L_x}, k_y = \frac{2\pi n_y}{L_y}$$



$$A = L_x L_y$$



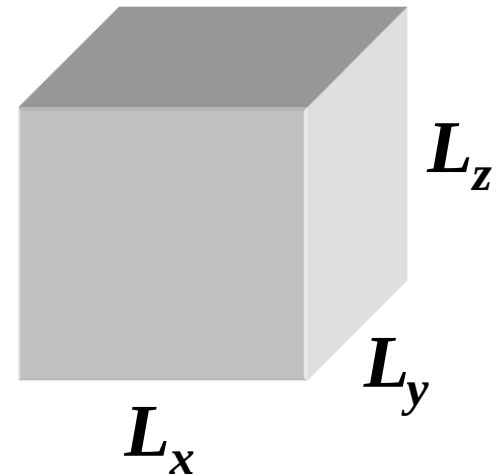
2D Fermi circle (费米圆)

$$\begin{aligned} N &= 2 \cdot \frac{\pi k_F^2}{\frac{2\pi}{L_x} \cdot \frac{2\pi}{L_y}} \\ &= 2 \cdot \pi k_F^2 \cdot \frac{A}{(2\pi)^2} \end{aligned}$$

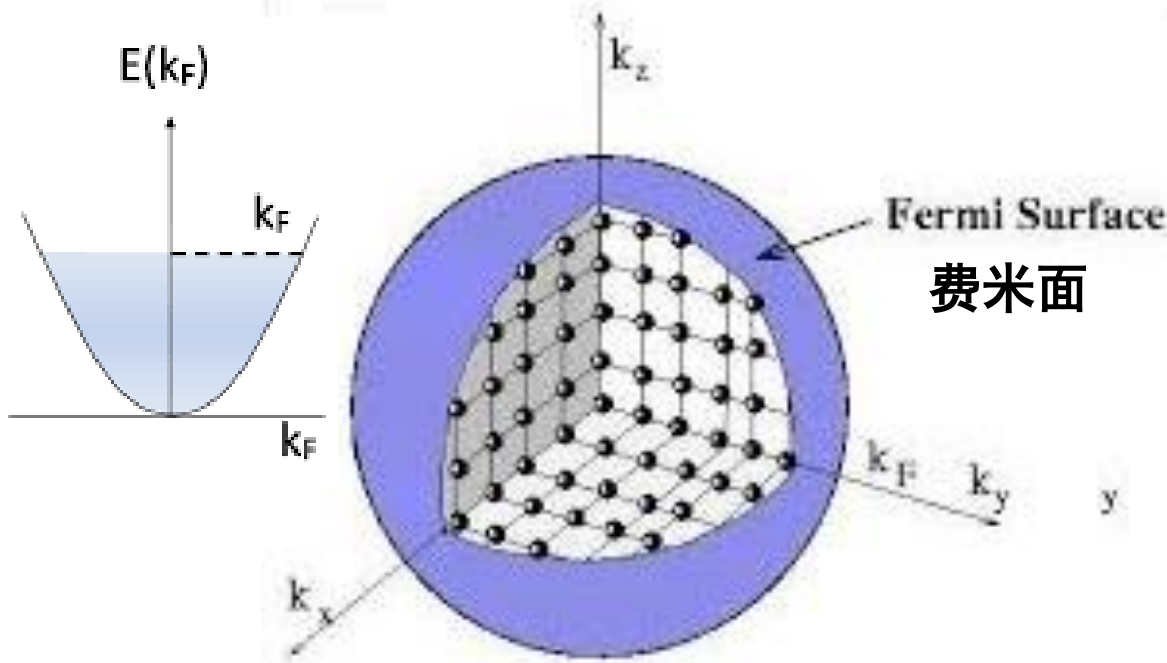
3D solid

in a 3D solid

$$k_x = \frac{2\pi n_x}{L_x}, k_y = \frac{2\pi n_y}{L_y}, k_z = \frac{2\pi n_z}{L_z}$$



$$V = L_x L_y L_z$$



3D Fermi sphere (费米球)

$$N = 2 \cdot \frac{\frac{4\pi}{3} k_F^3}{\frac{2\pi}{L_x} \cdot \frac{2\pi}{L_y} \cdot \frac{2\pi}{L_z}} = \frac{k_F^3}{3\pi^2} V$$

3D solid

$$\frac{N}{V} = \frac{k_F^3}{3\pi^2}$$

in a 3D solid

$n = N/V$ - free electron density

k_F - Fermi wavevector

E_F - Fermi energy

v_F - Fermi velocity

T_F - Fermi temperature

$$k_F = (3\pi^2 n)^{1/3}$$

$$v_F = \frac{\hbar k_F}{m_e}$$

$$E_F = \frac{\hbar^2 k_F^2}{2m_e}$$

$$T_F = \frac{E_F}{k_B}$$

These values are determined by density n , not N or V

highest *occupied* state at $T = 0$ K

Fermi surface

Density of States (DOS) 态密度

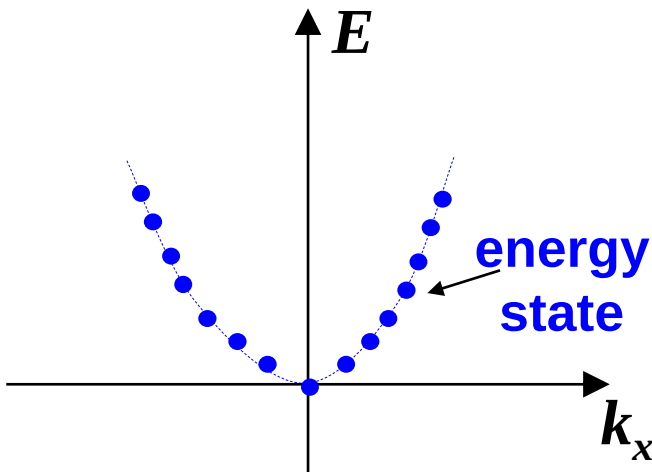
$$g(k)$$

DOS - number of energy states/levels in k space

Ashcroft & Mermin, p.35

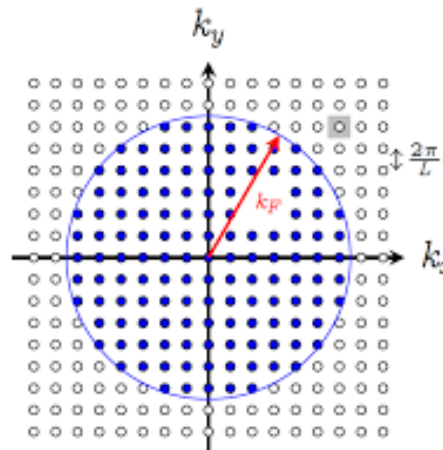
$$g(k) = 2 \frac{L}{2\pi}$$

1D



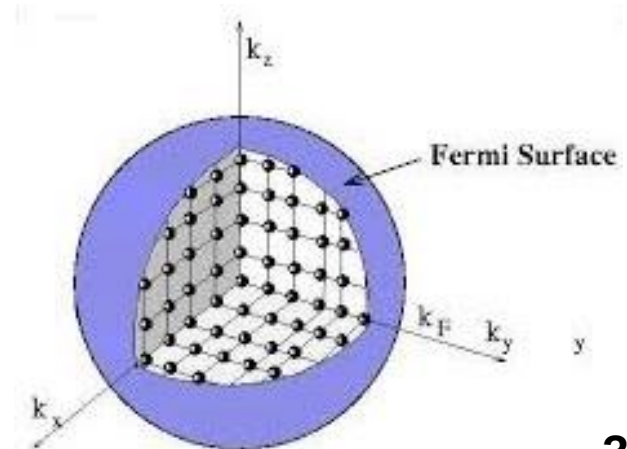
$$g(k) = 2 \frac{A}{(2\pi)^2}$$

2D



$$g(k) = 2 \frac{V}{(2\pi)^3}$$

3D



Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE}$$

DOS - number of energy states/levels per unit energy in $[E, E + dE]$, per unit volume

3D Solid

$$k = (3\pi^2 n)^{1/3}$$

$$E = \frac{\hbar^2 k^2}{2m_e}$$



$$n = \frac{1}{3\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{3/2}$$

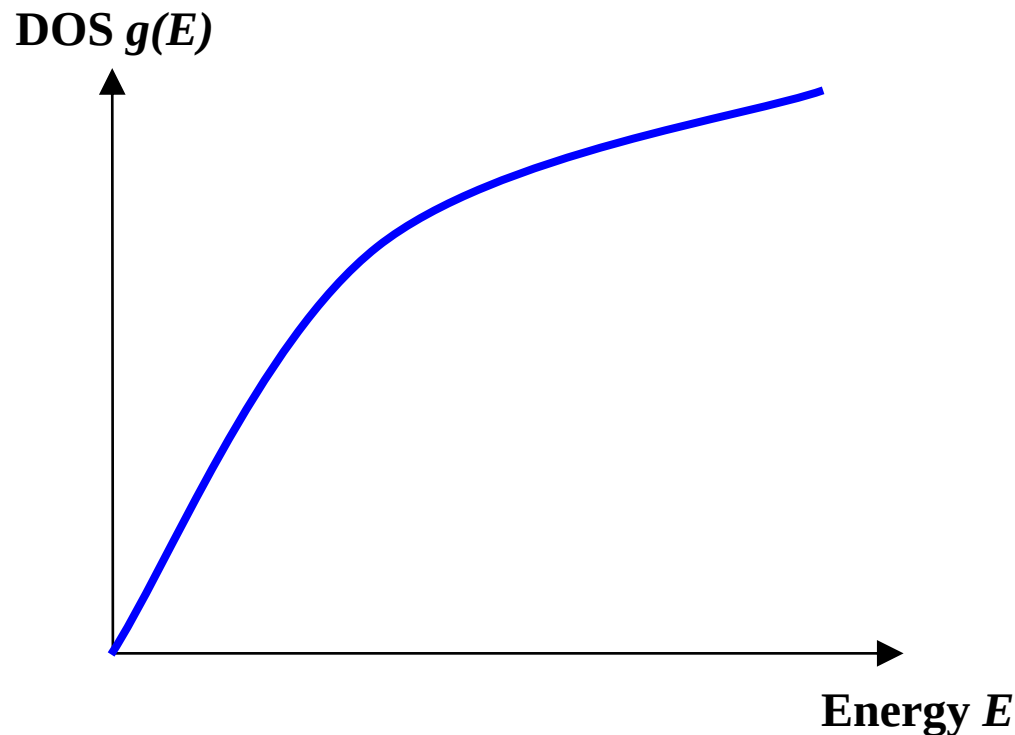


$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{1/2}$$

Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{1/2}$$

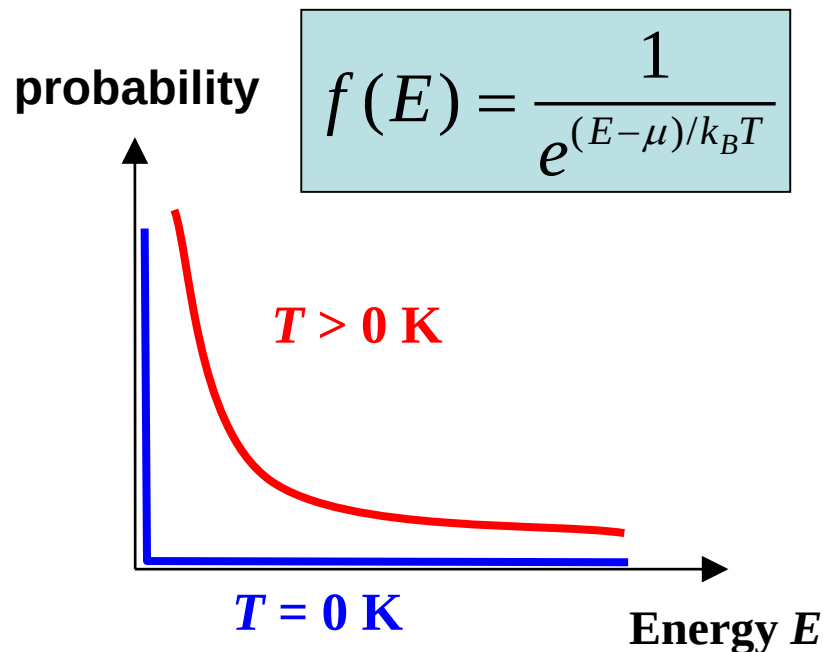
3D Solid



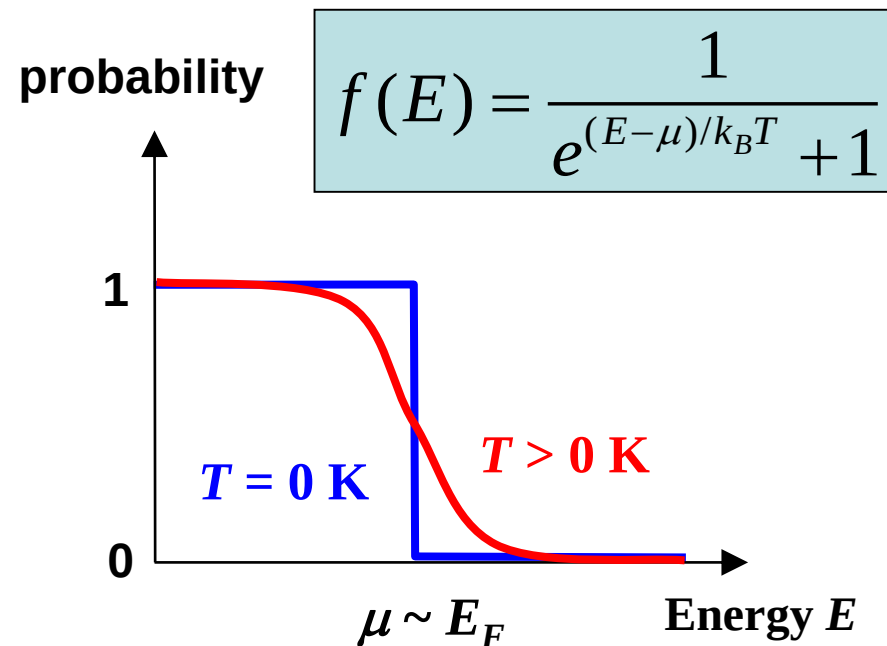
Q: How about in 1D and 2D?

Distribution of Free Electrons

Drude Model Maxwell–Boltzmann



Sommerfeld Model Fermi–Dirac



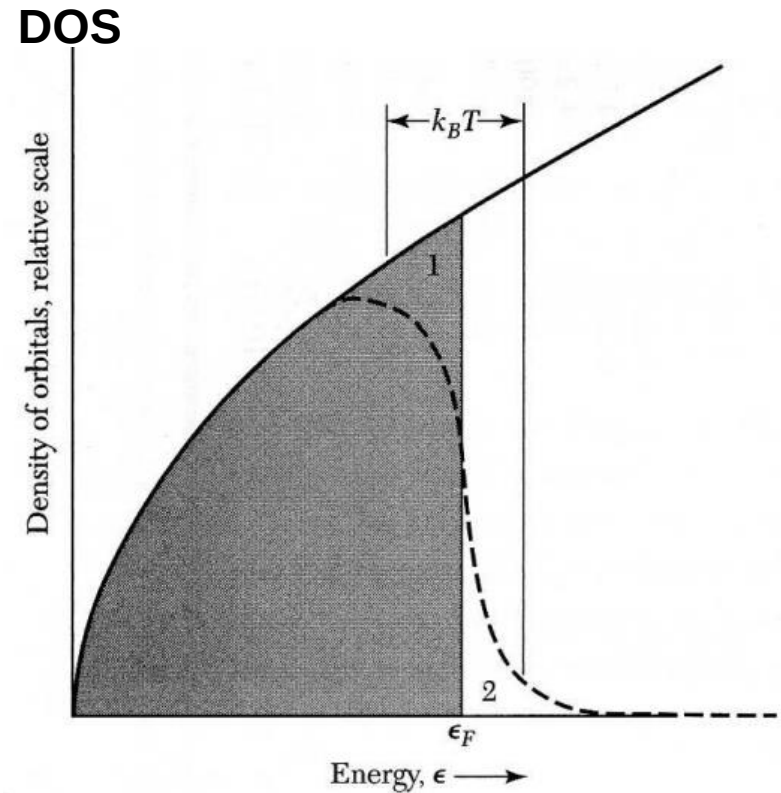
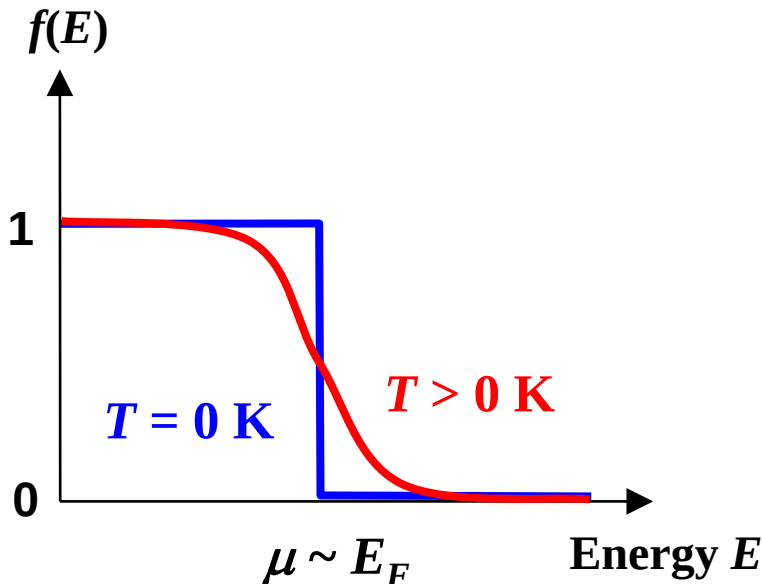
*Electrons occupy
higher levels of energy,
even at $T = 0 \text{ K}$*

Density of Electrons

Density of electrons = DOS * probability

$$g(E)f(E)$$

$$f(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}$$



When $T > 0 \text{ K}$, some electrons are excited to higher states (from 1 to 2)

State vs. Electron

energy state / level / orbital
能态 / 能级 / 轨道

electron / phonon / ...
电子 / 声子 / ...



determined by space, lattice,
environments, ...

Electrons in an Electric Field E

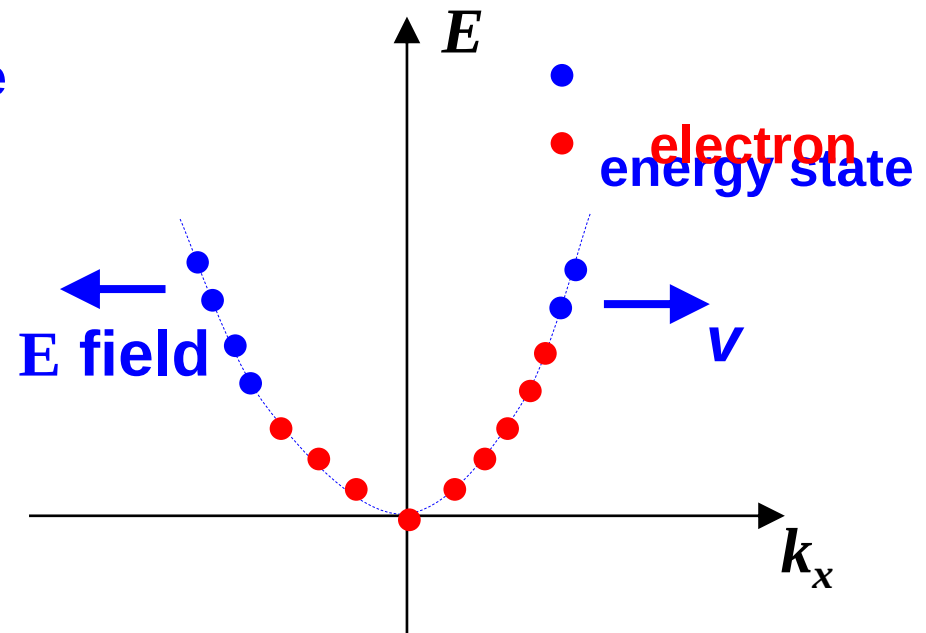
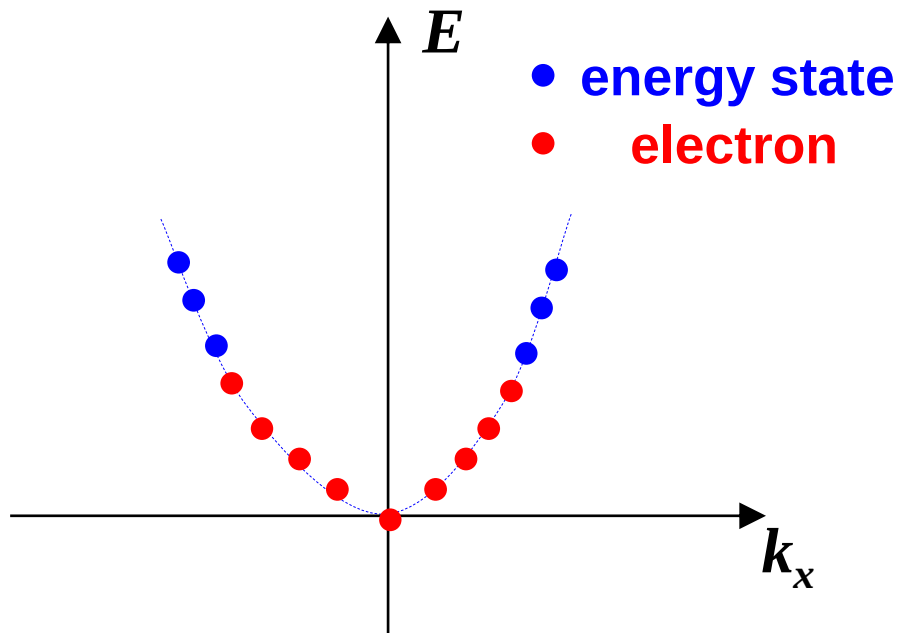
momentum 动量

$$p = mv = \hbar k$$

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt} = \hbar \frac{d\mathbf{k}}{dt} = -e\mathbf{E}$$



$$\delta \mathbf{k} = -\frac{e\mathbf{E}}{\hbar} t$$



Some electrons have high velocities/energies, even at $T = 0$ K

Electrons in an Electric Field E

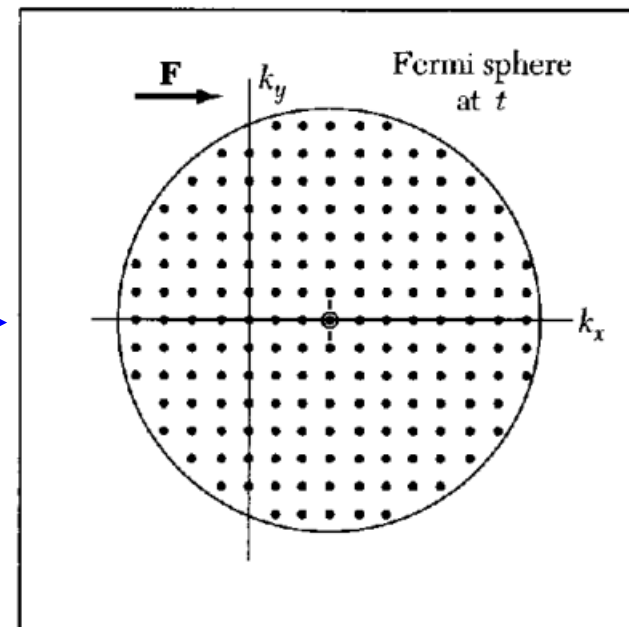
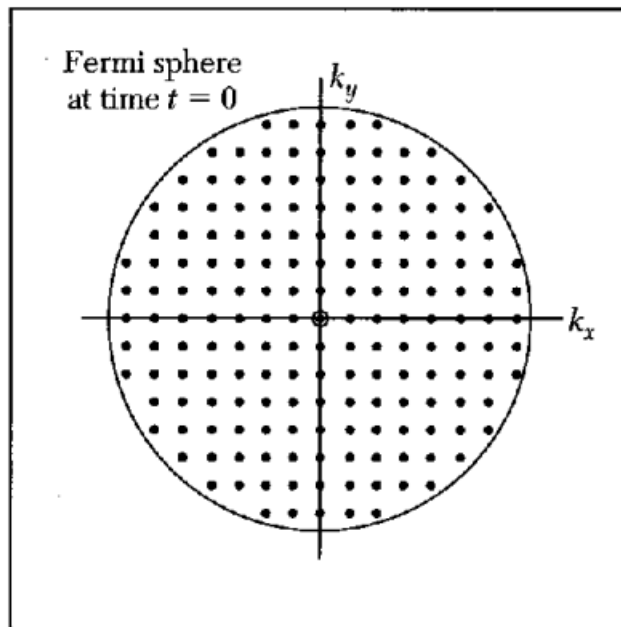
momentum 动量

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$$\delta \mathbf{k} = -\frac{e\mathbf{E}}{\hbar} t$$



Electrons in an Electric Field $\mathbf{E}$

momentum 动量

$$p = mv = \hbar k$$

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt} = \hbar \frac{d\mathbf{k}}{dt} = -e\mathbf{E}$$



$$\delta \mathbf{k} = -\frac{e\mathbf{E}}{\hbar} t$$

collision time $t = \tau$, the displacement $\delta \mathbf{k}$ is steady

$$\delta \mathbf{k} = -\frac{e\mathbf{E}}{\hbar} \tau$$



$$\mathbf{v} = \frac{\hbar \delta \mathbf{k}}{m} = -\frac{e\mathbf{E}}{m} \tau$$



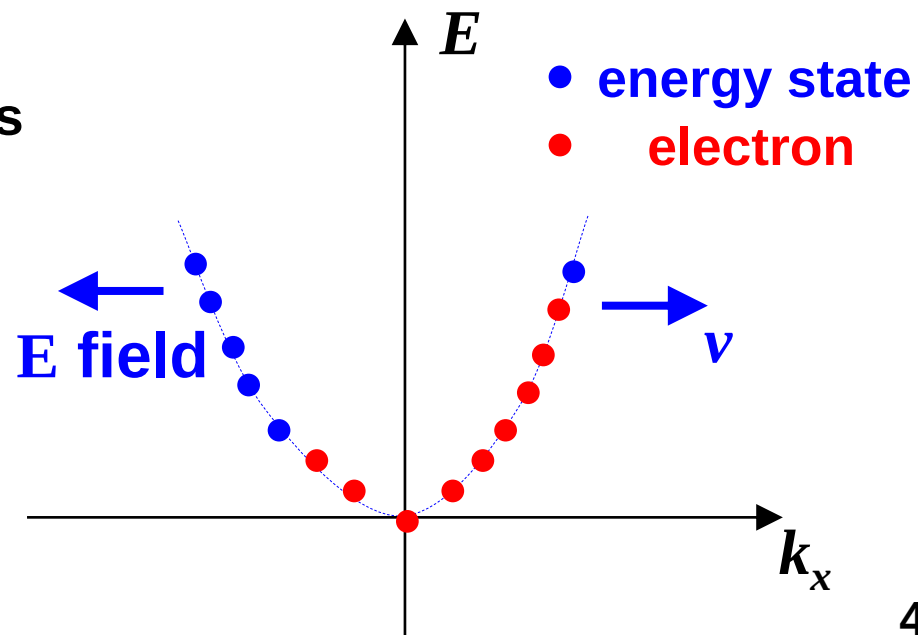
$$\mathbf{j} = -ne\mathbf{v} = \frac{ne^2\tau}{m} \mathbf{E} = \sigma \mathbf{E}$$

Ohm's law

Electron Conductivity - Revisit

- Electrons are in different energy states, therefore have different velocities and energies. Under the electric field, there are more electrons moving in the opposite direction.
- Mobility μ and relaxation time τ are average values for all the free electrons
- only σ is meaningful for metals

$$\sigma = ne\mu = \frac{ne^2\tau}{m}$$



Success of The Sommerfeld Model

- Ohm's Law
- Electronic conductivity σ
- Thermal conductivity of electrons
- Electronic heat capacity

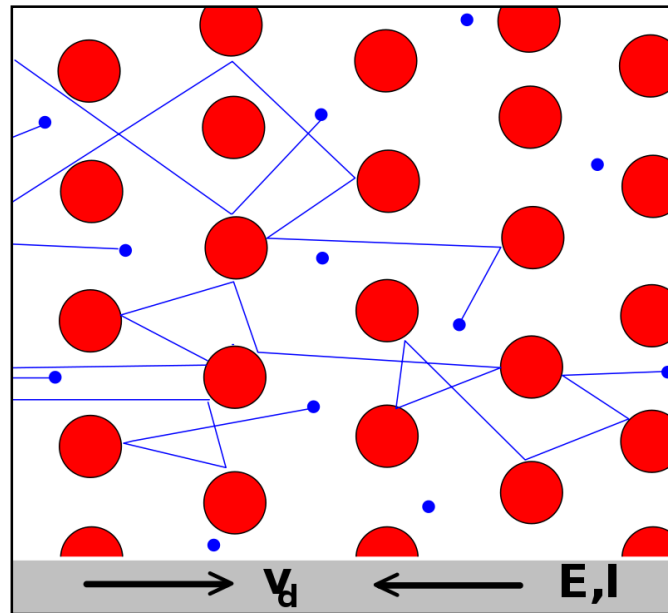
Failures of The Sommerfeld Model

It cannot explain

- **Electronic / Thermal properties of some other metals**
- **Hall effect / Hall coefficient**
- **Insulators / Semiconductors**
- **...**

Collision

- Both Drude and Sommerfeld assumed
 - Free electron
 - Resistance is caused by collision



What is the quantum mechanic view?

The Free Electron Models

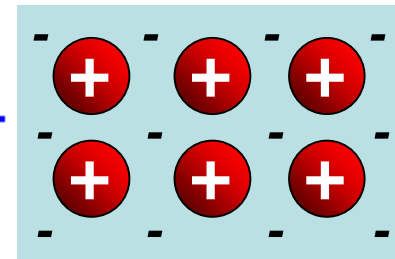
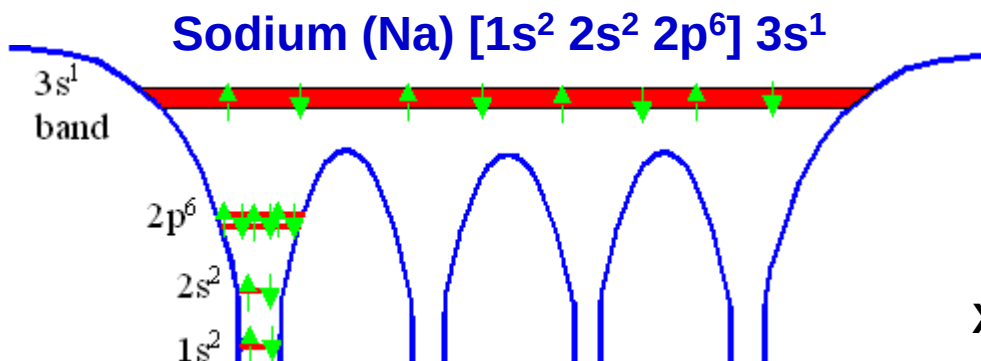
- The Drude Model: 1900s
- The Sommerfeld Model: 1920s
- What are missing?
 - ❑ Material and atom structures
 - ❑ Potentials of positive ions
 - ❑ Localized electrons
 - ❑ ...



P. Drude
1863–1906



A. Sommerfeld
1868–1951



positive ions
+
electron cloud

Next Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Thank you for your attention